



DEXHIGH SERVICES PVT LTD

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Junior Unity Developer — Technical Assessment

Dexhigh Services Pvt Ltd

Estimated effort: flexible | **Deadline:** 3 days (72 hours) from your confirmation

Objective

Build a small battle scene featuring two dragons — one controlled by the **player**, one controlled by **AI** — in a **2.5D top-down view**, similar to TFT or Dota Underlords.

This task evaluates your Unity fundamentals, your ability to implement game mechanics, and how effectively you use AI tools in your development workflow.

1. Scene Setup

- **Arena** — build an arena for the battle, with clear boundaries so neither dragon can leave the play area
- **Characters** — one player-controlled dragon and one AI-controlled dragon
- **Assets** — use free assets from the Unity Asset Store for models, animations and environment. Do not use paid or ripped assets
- **Camera** — 2.5D top-down perspective (angled overhead, like TFT or Dota Underlords), keeping both dragons in view during the fight

2. Player Controls

- **Movement** — WASD or click-to-move (your choice; state which in your README)
- **Abilities** — assign each ability to a key (for example 1 / 2 / 3, or Q / W / E)
- Controls should feel responsive, with no noticeable input delay

3. Combat Mechanics

Implement at least three distinct abilities:

- **Fire Attack** — ranged fire breath that deals damage in front of the dragon
- **Tail Attack** — close-range melee attack using the tail
- **Fly Attack** — the dragon takes off, attacks from the air, then lands

Each ability should have a damage value, a cooldown shown on its icon, and its own animation and visual effect.

4. Enemy AI

The AI dragon should fight back, not stand still. At minimum it should:

- Move towards or around the player
- Choose between its abilities based on distance (for example, tail attack when close, fire when at range)
- Respect the same cooldowns as the player

A simple state machine (Idle, Chase, Attack) is enough. Keep it readable.

5. Systems & UI

- **Health** — health bars for both dragons, clearly visible
- **Ability UI** — skill icons for all three abilities, with cooldown indicators
- **Hit feedback** — clear feedback when damage lands (flash, number popup, hit effect or similar)
- **Win condition** — when a dragon's health reaches zero, show a Winner Screen naming the victorious dragon, with a Restart button

6. Visual Quality & Polish

Make the battle as engaging as you can:

- Smooth animations for movement and attacks, with clean transitions
- Physics-based interactions where they add to the combat (knockback, impact)
- Fire particles and effects for the abilities
- Shadows and basic lighting in the arena
- Sound effects for attacks and hits

The build should run smoothly without major frame drops.

7. AI Usage Note (required)

At Dexhigh we value developers who use AI tools effectively. Include a short AI Usage Note covering:

- Which AI tools you used (e.g. Claude, ChatGPT, Gemini, GitHub Copilot)
- What you used them for — coding, debugging, research or planning
- One example where the AI got something wrong and how you fixed it
- How the tools made your work faster or better

Using AI is expected. We want to see that you understand and control the code you submit, not just paste it.

Reference

- **Camera angle and style:** <https://youtube.com/shorts/zayM2JLZeOU?si=19wblvKA3gQ4ULqO>
 - Please also research other auto-battler and top-down tactical games (TFT, Dota Underlords, etc.) for camera, UI and combat ideas
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Deliverables

1. **Source code** — GitHub repository link (preferred) or Google Drive link to the full Unity project
2. **Playable Windows build** — zipped, via Google Drive
3. **Gameplay video** — 1 to 2 minutes showing all three abilities, the AI fighting back, and the Winner Screen. Upload to YouTube as an unlisted video and share the link
4. **README** — Unity version used, controls, and the asset sources you used
5. **AI Usage Note** — in the README or as a separate document

Please make sure all links are set to "Anyone with the link can view".

What We Evaluate

Criteria	What good looks like
Working gameplay	All three abilities, AI, health and win condition function correctly
Code quality	Clean, readable C# with a sensible script structure
Project organisation	Tidy scene hierarchy and folders
Game feel	Responsive controls, smooth animation, clear feedback
Visual polish	Effects, lighting, shadows and sound
AI workflow	How thoughtfully you used AI tools

We are not expecting a finished game. A clean, fully working core loop scores better than lots of half-finished features.

Important

This is a free, unpaid assessment conducted only to evaluate your skills as part of our hiring process. Completing the test does not by itself guarantee selection — shortlisted candidates move on to an interview and our 30-day paid technical trial. Your submission will be used only to evaluate your application.

Submission

Send your deliverables to **sheikazarudeen8825@proton.me** within 3 days (72 hours) of confirming you have started.

Questions: **Sheik Azarudeen — HR, Dexhigh Services Pvt Ltd — WhatsApp only: 9360529479**